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Coupled management of electric vehicle workplace charging and office building loads

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ABSTRACT

Significant building energy savings are realized through heating, ventilation, and air conditioning (HVAC) setpoint adjustment and daylighting control. Workplace charging (WPC) enables colocation of electric vehicles (EVs) with office building loads. We proposed managing energy use of workplace EV charging and the office building and determined the number of EVs that building energy savings can facilitate charging. We simulated building energy savings in typical medium offices in Chicago IL, Baltimore MD, and Houston TX, spanning three US climate regions. Considering the EV hosting capacity of the saved building energy and travel patterns of roundtrip commuting, we minimized EV charging costs under time-of-use electricity pricing. Managed WPC can reduce charging electricity bills compared to first-come, first-served charging. The ratio of EVs to chargers, the coincident period of commuters' dwell time and lower electricity prices, and the number of EVs in the office impacted the economic benefits achieved through charging management.

1. Introduction

Charging infrastructure and management are determinants of the success of a smooth transition from gasoline to electric vehicles (EVs) (Li et al., 2020). Empirical evidence shows that charging is primarily performed at EV adopters' residences (Hardman et al., 2018; National Research Council, 2015; Traut et al., 2013), where vehicles are parked overnight. The residents' electricity bills can be minimized through efficient charging management (Ye et al., 2024). However, according to American Housing Survey (2017) data (US Census Bureau, 2017), only two-thirds of U.S. households can install chargers in residential garages or carports. The workplace, where a vehicle's dwell time on average reaches or exceeds 8 h, is the second most preferred charging location (Hardman et al., 2018; National Research Council, 2015; US Department of Energy, 2014). Workplace charging (WPC) could additionally provide charging opportunities for EV users who reside in multi-unit dwellings, are renting, and/or lack reliable home charging access (Huang and Zhou, 2015). WPC is found beneficial for employers, building managers, and employees (US Department of Energy, 2014). For both employers and building managers, chargers availability can attract employees and tenants and assists with accomplishing corporate sustainability goals tied to low carbon-intensity commuting. For employees, range anxiety can be mitigated with WPC access and flexibility. Even though the introduction of a small share of EVs is not projected to impact the electricity grid's performance (Hardman et al., 2018; Babrowski et al., 2014; Pearre et al., 2011; Schey et al., 2012), a greater number of EVs charging in workplaces offers not only opportunities to decarbonize work trips but also poses charging management challenges due to the coincident time of commuters' arrival to and departure from workplaces (Schey et al., 2012; Su and Chow, 2012). This leads to peak

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demand and transportation energy expenditure increase (Azadfar et al., 2015; Fetene et al., 2016). Therefore, using charging and other load management practices could help with peak shaving and/or valley filling, benefit the stability of the electric grid (or a workplace's microgrid), and avoid expensive electricity volumetric and demand charges (depending on the electricity rates offered by their electric utility) (National Research Council, 2015). To address the above challenges, managed WPC can help with realizing colocation of EV and office building energy use benefits via load shifts and optimization.

Workplace EV charging is often treated as a separate load in the existing literature, without identifying opportunities for concurrent management of office buildings and the parking lot-associated loads (Huang and Zhou, 2015; Nguyen and Le, 2013; Le et al., 2015). However, there is empirical evidence that uncovers WPC financial feasibility (Tulpule et al., 2013; Williams and DeShazo, 2014), as well as studies that explore the relationship of WPC and EV adoption (Xie et al., 2018) and evaluate incentives to improve the acceptance of WPC (Fetene et al., 2016). Integrating EVs with office buildings load management can enable flexible energy use with smart EV charging strategies (Zhou et al., 2019). To guide planning and operations of WPC, optimization frameworks are proposed to satisfy EV charging demand, considering the power level of chargers, installation and operation expenses, and other operating constraints (e.g., charging rates, user preferences) (Huang and Zhou, 2015). Other optimization models maximize EV drivers' miles, accounting for WPC budget constraints (Li et al., 2020). Modeling assumptions include that once an EV connects to a workplace charger, it will not disconnect until leaving the parking lot (Li et al., 2020). Others develop a bi-level programming model to optimize EV charging schedules, considering a limited number of chargers in the workplace (Liu et al., 2020), while others integrate single family residential building, office, and EVs energy systems (Barone et al., 2019), by simulating the coupling of photovoltaic panels and EV battery swapping options management; they find that EVs can effectively benefit such integrated systems' operation by saving energy.

In this study, we aim to uncover management benefits of the EV-office building nexus, afforded by contemporary office building energy control and EV charging sessions optimization, for energy efficiency and low electricity bills. This research contributes to filling research gaps in coupled management of workplace building energy use and EV charging schedules. We augment existing knowledge as follows:

- We investigated for the first time the impact of office building control interventions, like adjustments to the heating, ventilation, and air conditioning (HVAC) setpoints and the implementation of daylighting control, on accommodating the workplace EV charging demand and schedule.
- We proposed an optimization model to minimize EV charging costs at offices, considering limited availability of chargers and sharing, time-of-use electricity pricing with and without demand charges, commuting patterns, and potential energy savings in office buildings.
- We estimated the impact of workplace EV charging management on electricity bills. By implementing coupled EV and office building energy management, we maintained energy consumption of the building to the same level as before the advent of EVs in the office parking lots, while minimizing electricity bills.

We simulate representative office building energy consumption, under two building control strategies, i.e., thermostat setpoint adjustments and daylighting control, to estimate building energy savings. Our simulations model the time-of-use power profiles of the building without EV charging. Then, we estimate WPC demand that is accommodated through the office HVAC setpoint or lighting adjustments, so as to keep the total energy use from significantly increasing as more EVs are colocated in the office's parking lot. For instance, by adjusting the HVAC setpoint the saved energy can be used to recharge the EVs at the workplace. We analyze EV commuting patterns to estimate the total kWh that need to be recharged in the workplace, the distance traveled from home to workplace, and the arrival times to and departure times from the workplace (Zhang et al., 2014, 2020; Li et al., 2020), using the National Household Travel Survey (NHTS) 2017 data (United States Department of Transportation, 2017). We formulate an optimization model that minimizes EV charging costs through WPC sessions management, responsive to time-of-use volumetric electricity tariffs for an aggregator or building manager (US Department of Energy, 2014; Huang and Zhou, 2015). When EVs recharge at workplaces, the manager needs to consider the EVs demand and the baseload energy consumption of the building (e.g., HVAC and lighting) (Laustsen, 2008). If the shedding of the office's energy use can satisfy the EV WPC demand and the total costs (i.e., building and EV charging electricity bill) under the managed EV charging schedule are minimized, then the coordinated building and EV charging loads management would be both energy efficient and economical. Fig. 1 presents the workflow of this research. Due to climate differences in the US (Karl and Koss, 1984) that impact building energy consumption (Huang and Gurney, 2016) and charging demand (Gilleran et al., 2021), modeling regional office energy use and travel patterns is essential. Our analysis focuses on three climate regions (i.e., central, northeast, and south) with distinct climates and spatiotemporal transportation and energy use patterns.

The remainder of this paper is organized as follows. Section 2 presents the office building control strategies and the optimization model for EV WPC management. Section 3 introduces the input parameters of the optimization model and the case study settings. In addition, the results of implementing office building control strategies and EV charging management are shown. Finally, Section 4 concludes with discussion of main findings, limitations of this research, and future research directions.

2. Methodology

2.1. Office building energy consumption simulation

The office building energy consumption simulations are performed in EnergyPlus (Crawley et al., 2000), a whole-building energy modeling software. EnergyPlus obtains (a) building construction specs, such as building geometry and materials (e.g., walls, floors,

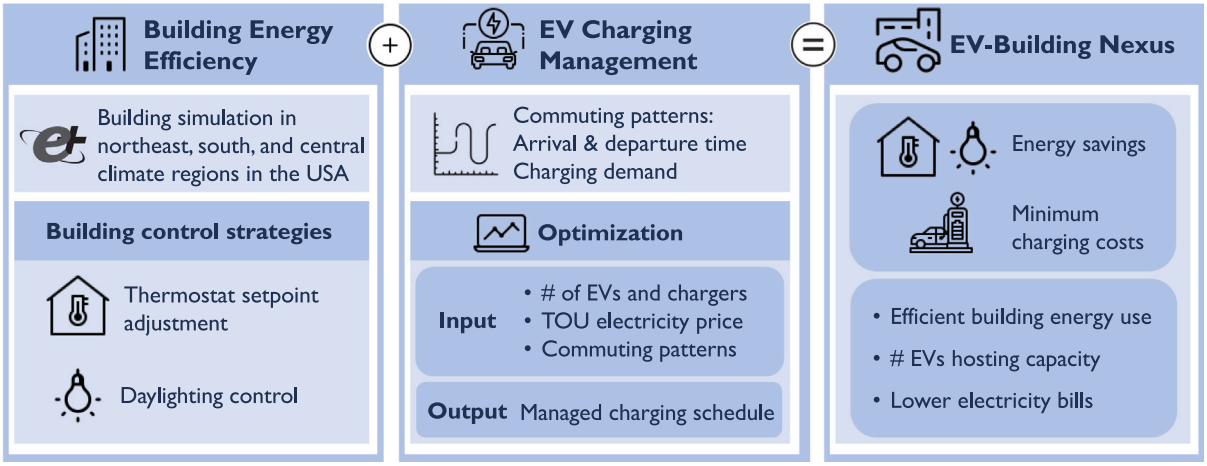


Fig. 1. A schematic of the research approach followed for managing the electric vehicles workplace charging and office building energy use (EV-office building nexus) in representative US climate regions (i.e., northeast, south, and central).

ceiling, windows, etc.), (b) building use information, such as number of occupants and their occupancy schedules, and (c) HVAC and lighting operation schedules to simulate its energy use. Furthermore, EnergyPlus uses climate information for the city where the building is situated at. The weather data stem from a typical meteorological year for each simulated city and contain annual hourly information on median weather conditions over a multiyear period (Wilcox and Marion, 2008).

The US DOE has funded the development of commercial reference buildings, which correspond to 16 building types, ranging from schools, hospitals, and warehouses to office buildings, which represent approximately 70% of the US commercial buildings (Deru et al., 2011). These models are used as a baseline of comparison in studies related to buildings' energy efficiency and controls. In our study, we model energy use of the US DOE medium office reference building: each floor has four perimeter zones and one core zone; both cooling and heating use electricity as the source of energy; the window-to-wall ratio is 33%; and the total floor area is 5,000 m^2 (53,600 ft^2). This building uses a cooling setpoint of 24 °C and a heating setpoint of 21 °C in all US climates and does not implement daylighting controls. We simulated that reference building and used it as the baseline for our study. Then, we applied two separate control strategies as follows:

- Control strategy 1: Adjust the building's thermostat setpoint from 24 to 26 °C with a discrete step of 0.5 °C during the summer period (June–August (National Centers for Environmental Information, 2016)); adjust the building's thermostat setpoint from 21 to 19 °C with a discrete step of 0.5 °C during the winter period (December–February (National Centers for Environmental Information, 2016)). Adjusting the thermostat setpoint within these margins does not significantly affect occupant thermal comfort levels (American Society of Heating, Refrigerating, and Air-Conditioning Engineers, 2017) and leads to energy savings.
- Control strategy 2: Implement daylighting control to the four perimeter offices of each floor. The daylighting control calculates the contribution of daylight and dims the electric lighting to meet a target illuminance level in the office. We used two distinct illuminance levels of 300 and 500 lux (US General Services Administration, 2019) at a work plane height of 0.76 m.

Results from the office building energy consumption simulations are derived for summer and winter seasons in three cities: Chicago, IL; Baltimore, MD; and Houston, TX. The three cities and various simulated scenario energy use outcomes are presented in Table 1. These represent three distinct humid climate types in the United States, with Chicago characterized by a cold and humid climate, Baltimore by a mild and humid climate, and Houston by a hot and humid climate (American Society of Heating, Refrigerating, and Air-Conditioning Engineers, 2013). Our analysis takes into account only workdays, excluding holidays and weekends, and considers times of day when at least 40% of the building occupants are present (i.e., excluding late evenings, night time) and therefore can charge their EVs by leveraging energy savings from the building's controls. The building energy simulations show that raising the cooling setpoint in the summer and decreasing the heating setpoint in the winter lead to major energy savings. Energy savings are higher for an office building in a hot and humid climate during the summer period and less during the winter period, since outdoor temperature and humidity conditions drive the HVAC energy consumption. Also, the daylighting control leads to energy savings that vary across climate regions, because the amount of solar radiation through the windows depends on the latitude of the building's location. During summer and winter, the Houston TX sun is mostly overhead, whereas at Baltimore MD and Chicago IL the solar radiation enters the perimeter offices because of the lower solar angle since both cities are located northern in the hemisphere.

2.2. Electric vehicle workplace charging management

We formulated a mathematical model to determine the WPC schedule of EVs, which are colocated with the medium office building. We learned from existing charging scheduling optimization frameworks in the literature (Li et al., 2020; Huang and Zhou,

Table 1
Office building energy use savings in three US climate regions under various HVAC thermostat setpoint adjustment and daylighting control strategies.

#	City, State	Climate type	Control strategies	Summer energy savings (kWh/day)	Winter energy savings (kWh/day)
Baseline	Chicago, IL	Cold, humid	–	–	–
Scenario 1a	"	"	HVAC setpoint ΔT by 0.5 °C	47	74
Scenario 1b	"	"	HVAC setpoint ΔT by 1 °C	89	145
Scenario 1c	"	"	HVAC setpoint ΔT by 1.5 °C	120	214
Scenario 1d	"	"	HVAC setpoint ΔT by 2 °C	141	279
Scenario 1e	"	"	daylighting control at 300 lux	281	175
Scenario 1f	"	"	daylighting control at 500 lux	240	145
Baseline	Baltimore, MD	Mild, humid	–	–	–
Scenario 2a	"	"	HVAC setpoint ΔT by 0.5 °C	52	72
Scenario 2b	"	"	HVAC setpoint ΔT by 1 °C	100	141
Scenario 2c	"	"	HVAC setpoint ΔT by 1.5 °C	138	205
Scenario 2d	"	"	HVAC setpoint ΔT by 2 °C	164	264
Scenario 2e	"	"	daylighting control at 300 lux	273	178
Scenario 2f	"	"	daylighting control at 500 lux	231	145
Baseline	Houston, TX	Hot, humid	–	–	–
Scenario 3a	"	"	HVAC setpoint ΔT by 0.5 °C	82	59
Scenario 3b	"	"	HVAC setpoint ΔT by 1 °C	155	110
Scenario 3c	"	"	HVAC setpoint ΔT by 1.5 °C	216	154
Scenario 3d	"	"	HVAC setpoint ΔT by 2 °C	265	190
Scenario 3e	"	"	daylighting control at 300 lux	194	146
Scenario 3f	"	"	daylighting control at 500 lux	119	106

2015; Liu et al., 2020) and modified them as follows. Huang and Zhou aimed to minimize total costs of equipment, installation, and operation of charging stations in an office building to meet employees' required charging demand (Huang and Zhou, 2015). We set a different objective, aligned with minimizing charging costs for the office manager, to determine the EV charging schedule. Compared to other charging scheduling models (Li et al., 2020), where the optimization objective is maximizing the satisfied charging demand, our model ensures that the heterogeneous charging demands are met as constraints. Our model allows drivers or office parking attendants to move EVs at the end of their charging session to enable shared use of the limited office lot charging infrastructure. When several EVs are allocated to recharge at the same charger, the dwell time of these EVs can overlap. A vital task of the charging management is to mitigate these overlaps among charging sessions allocated to the same charger. Beyond the overlap scenarios of charging sessions proposed by Liu et al. (2020), we significantly modified the model to account for all EV charging sequences and now fully capture the complete set of overlapping scenarios, as described in the next subsection.

2.2.1. Electric vehicle allocation to workplace chargers

We defined a single day as the management period for the charging sessions in the parking lot of a medium-sized office. The day is divided into time increments, represented by t (where $t = 1, 2, \ldots, T$), with each increment spanning p, i.e., ten minutes. We account for commuting travel patterns when setting the recharging energy per EV n as D_n, the arrival time at the office t_n^a, the departure time from the office t_n^d, and dwell time t_n^{dwell} at the office's parking lot. Those are parameters of the proposed WPC scheduling model, where:

$$t_n^{dwell} = (t_n^d - t_n^a) \times p. \tag{1}$$

To solve the EVs charging session management problem, the sequence of the charging sessions at each WPC charger needs to be determined. We proposed a heuristic process to assign scores to each EV based on the charging priority related to arrival time, departure time, dwell time, and charging demand. EVs that arrive earlier, depart earlier, dwell shorter, and require more energy are scored higher. EVs with a higher score will be charged first. When EVs have the same scores, the EV arriving earlier has charging priority. If several EVs have the same score and arrival time, the EV with the earlier departure time will be charged first because of a more urgent charging request. If the score, arrival time, and departure time are the same for several EVs, the EV requesting more energy will have WPC priority. Therefore, the charging sequence by sorting the EVs can be defined as EV = $\{EV_1, EV_2, \ldots, EV_N\}$.

After determining charging priority, the overlapping cases of EVs allocated to recharge at one particular charger need to be analyzed. This step influences the time allocation of charging sessions in each charger, as Fig. 2 shows. If EV n and EV m are allocated to recharge at the same charger, and EV n has charging priority, six overlapping cases are plausible. We define t_n^a and t_m^a as the arrival times of the aforementioned vehicles, t_n^d and t_m^d as the departure times, t_n^e as the time to stop charging for EV n, and t_m^s is the time to start charging for EV m. For instance, in Cases 1 and 2 we assume that the arrival time of EV n is earlier than EV m, but EV n departs from the office parking lot earlier in Case 1 and later in Case 2. Thus, the overlap period is $[t_m^a, t_n^d]$ in Case 1 and $[t_m^a, t_m^d]$ in Case 2. During $[t_n^a, t_n^e]$ the charger serves EV n, while it serves EV m during $[t_m^s, t_m^d]$. To share the charger efficiently, the end time to serve EV n should be the same time point as the start time to serve EV m. Thus, t_n^e and t_m^s are the time point for the charger to move from charging EV n to EV m, which is the critical time point that the conventional optimization charging schedule methods search for. In fact, when we guarantee that the required charging demands are satisfied during EV dwell times, the exact t_n^e and t_m^s determination can be neglected. Therefore, we only use two conditions to present the Case 1 in Fig. 2, which is the only case

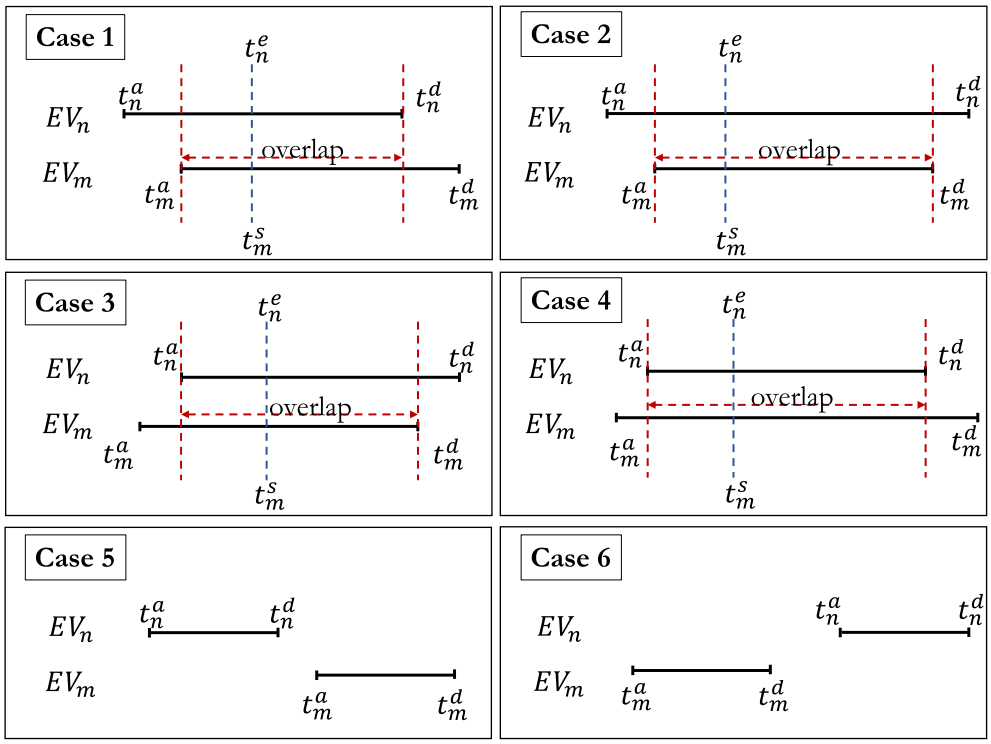


Fig. 2. Overlapping cases for EV dwell times assigned to a shared workplace EV charger.

considered by Liu et al. (2020), as follows: (1) the time from t_n^a to t_m^a is not enough for EV n to satisfy their charging demand; (2) the time from t_n^d to t_m^d is not enough for EV m to satisfy their charging demand. However, in Case 2, the constraints are different. We need to ensure that the charging power during $[t_n^a, t_m^a]$ cannot exceed the required energy of EV n, and the summation of charging power during $[t_n^a, t_m^d]$ must satisfy the charging demand of EVs n and m.

In an unmanaged charging case, a first-come, first-served (FCFS) charging practice is modeled. EVs find an unoccupied charger and the earliest arriving EV has priority to charge. If all workplace chargers are occupied, the subsequent arriving EV will wait in a virtual queue for the next available charger.

In both managed and unmanaged charging cases, we assume that if an EV n and the following EV m share the same charger i, and EV n completes recharging at time increment t, the energy drawn from the charger to EV n is $P_{ev(n,t)}$. Thus, the energy that EV m can utilize is $P_{max} - P_{ev(n,t)}$, where P_{max} is the maximum output power of the charger in an hour.

We formulate our problem by setting as our objective the minimization of the system's charging costs responsive to time-of-use volumetric electricity rates, incurred by the building manager and/or corporation, and determine the sequence of charging sessions on workplace chargers.

2.2.2. Electric vehicle workplace charging management formulation

Given a workplace's office parking lot, we schedule EV charging to minimize charging costs during a typical summer workday. The model needs four sets of inputs as follows:

- Charging patterns: This set of data includes the arrival and departure times, and the required charging demand of the commuters who recharge their EVs in the workplace. The number of EVs is predetermined and calculated based on the energy saved by building efficiency improvements.
- Charger allocation: EVs need to share the limited number of chargers, while each EV is assigned to a specific charger to recharge. The number of chargers is determined by observing the EV charging patterns. We computed the minimum number of shared chargers required to satisfy the charging demand of all the EVs in the workplace parking lot.
- Electricity price: The time-of-use electricity volumetric pricing scheme stems from each city's electric utility and varies by season. In WPC management, we only consider the energy charge, excluding the demand charge in the optimization process. That is because the demand portion of the electricity bill does not vary by time of day. We add the demand charge in the electricity bill to comprehensively assess the savings incurred from our management approach.
- Charger specification: This includes the type of chargers and their technical specifications (e.g., charging power).

Table 2
Nomenclature.

Indices			
i	index of chargers, $i \in I$	n	index of EVs, $n \in N$
t	index of time increments, in a 24-h day, $t \in T$	m	index of EVs to differentiate from n, $m \in N$
Parameters			
t_n^a	arrival time at the workplace of an EV n	t_n^d	departure time from workplace of an EV n
P_{max}	maximum charging power in a time increment (kWh)	D_n	required energy for an EV n to complete the round commute trips (kWh)
$Cost(t)$	time-of-use electricity price for energy consumption (based on the volumetric rate $/kWh)	$\delta_{i,n}$	equals to 1 if EV n charges at charger i; otherwise 0
$L_{i,n}$	charging position of EV n at charger i		
Decision Variable			
$y_{i,t}$	output energy of charger i in time increment t (kWh)		

In the charging management model with charging infrastructure sharing, the allocation of a charger to each EV is defined by δ $(I \times N)$, as follows:

$$\delta_{i,n} = \begin{cases} 1, & \text{if EV } n \text{ charges at charger } i \\ 0, & \text{otherwise} \end{cases} \quad \forall n \in N, i \in I \tag{2}$$

Each EV will be assigned to a unique charging station until its charging demand is satisfied or the departure time is reached. This notion is captured by the constraints of δ:

$$\sum_{i=1}^{I} \delta_{i,n} = 1, \forall n \in N \tag{3}$$

When several EVs are assigned to the same charger, based on the charging sequence obtained by the model, we know the charging position of each EV in this specific charger, denoted as L. L presents the indices of charging sessions corresponding to each EV and the charger.

$$L_{i,m} = \sum_{n=1}^{m} \delta_{i,n}, \text{ if } \delta_{i,m} = 1 \tag{4}$$

Given the sets of inputs, we construct the model in Eqs. (5)–(13), with the notation of indices, parameters, and variables in Table 2:

$$\min \sum_{i \in I} \sum_{t \in T} Cost(t) \cdot y_{i,t} \tag{5}$$

s.t.

$$\begin{cases} \sum_{t=t_n^a}^{t_m^a} y_{i,t} \le D_n \\ \sum_{t=t_n^d}^{t_m^d} y_{i,t} \le D_m \end{cases}, \text{if } t_n^a \le t_m^a \ \&\ t_n^d \le t_m^d \ \&\ L_{i,n} + 1 = L_{i,m}, \forall i \in I \tag{6}$$

$$\begin{cases} \sum_{t=t_n^a}^{t_m^a} y_{i,t} \le D_n \\ \sum_{t=t_n^a}^{t_m^d} y_{i,t} \ge D_n + D_m \end{cases}, \text{if } t_n^a \le t_m^a \ \&\ t_n^d \ge t_m^d \ \&\ L_{i,n} + 1 = L_{i,m}, \forall i \in I \tag{7}$$

$$\begin{cases} \sum_{t=t_n^a}^{t_n^d} y_{i,t} \ge D_n \\ \sum_{t=t_n^d}^{t_m^d} y_{i,t} \le D_m \end{cases}, \text{if } t_n^a \ge t_m^a \ \&\ t_n^d \le t_m^d \ \&\ L_{i,n} + 1 = L_{i,m}, \forall i \in I \tag{8}$$

$$\sum_{t=t_n^a}^{t_m^d} y_{i,t} \ge D_n + D_m, \text{if } t_n^a \ge t_m^a \ \&\ t_n^d \ge t_m^d \ \&\ L_{i,n} + 1 = L_{i,m}, \forall i \in I \tag{9}$$

Table 3
Typical commuting patterns in Chicago IL, Baltimore MD, and Houston TX.

	Chicago, IL	Baltimore, MD	Houston, TX
Workplace arrival time	8:00	8:00	8:00
Workplace departure time	16:30	17:00	17:00
Roundtrip commuting distance (miles/day)	25.01	26.30	25.18
Required WPC energy (kWh/day)	6.71	7.06	6.76

$$\begin{cases} \sum_{t=t_n^a}^{t_n^d} y_{i,t} = D_n \\ \qquad\qquad , \text{if } [t_n^a, t_n^d] \cap [t_m^a, t_m^d] = \emptyset \ \& \ L_{i,n} + 1 = L_{i,m}, \forall i \in I \\ \sum_{t=t_m^a}^{t_m^d} y_{i,t} = D_m \end{cases} \tag{10}$$

$$y_{i,t} = 0, \quad \forall t \notin [t_n^a, t_n^d], i \in I, n \in N, \text{if } \delta_{i,n} = 1 \tag{11}$$

$$\sum_{t=t_n^a}^{t_n^d} y_{i,t} = D_n, \text{if } \delta_{i,n} = 1, i \in I, n \in N \tag{12}$$

$$0 \le y_{i,t} \le P_{max}, \quad \forall i \in I, t \in T \tag{13}$$

The objective function (5) minimizes the total charging costs, based on the volumetric energy charge that varies per time of day during a typical workday. Constraints in Eq. (6) to Eq. (10) indicate the overlapping EV scheduling cases (i.e., Case 1 to Case 5), as shown in Fig. 2. Constraints in Eq. (11) demonstrate that EVs cannot recharge beyond their dwell time in the workplace. The constraints in Eq. (12) determine that each EV charging demand needs to be satisfied. The charging power drawn from the grid $y_{i,t}$ to recharge each EV's battery has upper and lower bounds that should not be violated during each time interval, as shown in constraints of Eq. (13).

According to the algorithm proposed by Liu et al. (2020), the mathematical model can be broken down in two parts: determination of the allocation matrix δ, and the linear program to optimize the charging schedule. The first can be solved by the combination of genetic algorithms (Rocke, 2000) and the linear program can be directly solved by a commercial solver. We initially constructed a charger assignment matrix δ following the allocation constraints shown in Eq. (3), enforcing each EV to be assigned to recharge at only one charger. Using the assignment matrix δ along with charging patterns, charger specification, and electricity price as the input, the mathematical model becomes a linear programming problem. We use Gurobi Optimizer version 9.5.1 (Gurobi Optimization, LLC, 2022) to solve the linear programming problem and obtain the optimal results and corresponding assignment matrix in each generation. The assignment matrices with the smallest objective value in each generation can produce offspring through crossover and mutation. Then, we use the new generation of the assignment matrix and solve the linear programming until the predetermined generation is reached. Finally, the algorithm can output the charging power matrix and the corresponding charger's assignment matrix δ.

3. Numerical experiments

We conducted numerical examples in three populous US cities, leveraging their spatiotemporal travel patterns and charging demand from the 2017 NHTS (United States Department of Transportation, 2017), modeled office building energy consumption, and accounted for contemporary charging technology, and time-of-use electricity prices of their electric utilities. The detailed data preparation processes are presented in the Section 3.1 followed by the results and discussion in the rest of this Section 3.

3.1. Data preparation

EV charging demand is estimated based on NHTS commuting travel patterns (United States Department of Transportation, 2017). The roundtrip commuting distance is assumed to correspond to the workplace charging demand. The probability distribution of office arrival and departure times also stem from the NHTS, as presented in Fig. 3. Based on these distributions, we generated one hundred synthetic data samples using Monte Carlo simulation, including arrival and departure times of EVs to and from the office building, and roundtrip commuting distances of their drivers. Our computation demonstrates that energy savings from adjusted HVAC setpoints and lighting in office buildings can support EV travel demand in Chicago, IL, Baltimore, MD, and Houston, TX. A typical sample for each of the cities was also created, with the most frequent arrival and departure times to and from the office, and the roundtrip commuting distance, presented in Table 3. We assume that the EV is a Nissan Leaf (USA, 2022) vehicle, one of the most popular and affordable US EV models, with efficiency of 3.725 miles/kWh. Therefore, we can infer the required charging demand distribution in the synthetic data samples and typical samples.

In the rest of this Section 3, we computed the daily amount of energy saved (kWh) by the HVAC thermostat setpoint adjustments and daylighting controls at the office building in these 3 climate regions. Given that we already have an estimate of the charging demand of the typical samples, we can compute the number of EVs that can be supported by the saved energy from the building

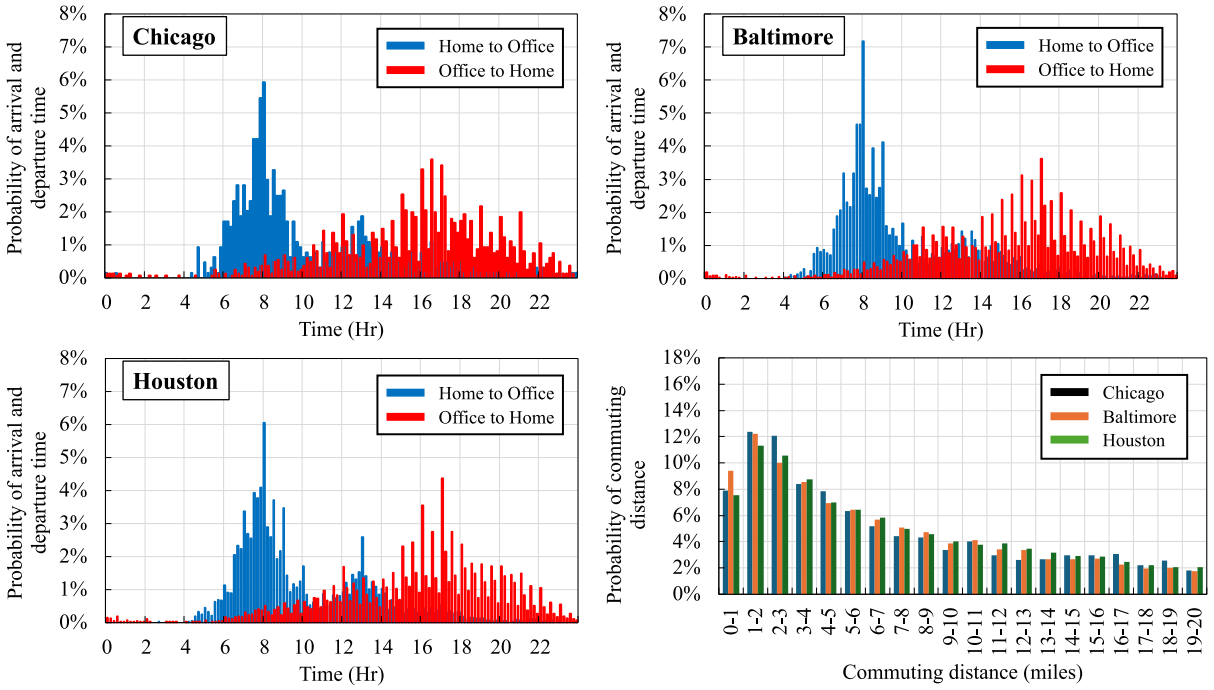


Fig. 3. Distribution of arrival times at the office and departure times from the workplace, and travel distance for commuters in Chicago IL, Baltimore MD, and Houston TX.

Table 4
The number of EVs and corresponding year of EV adoption supported by the energy savings from the HVAC setpoint adjustments and daylighting control in Chicago IL, Baltimore MD, and Houston TX, based on typical commuting patterns.

		Chicago, IL		Baltimore, MD		Houston, TX	
		#EVs	Year	#EVs	Year	#EVs	Year
HVAC setpoint ΔT (°C)	+0.5	7	2024	8	2022	12	2023
	+1	13	2026	15	2024	23	2024–2025
	+1.5	18	2027	20	2024–2025	32	2025–2026
	+2	21	2027–2028	24	2025–2026	39	2026–2027
Daylighting control (lux)	500	36	2029–2030	34	2026–2027	18	2024
	300	42	2030	40	2027–2028	29	2025–2026

efficiency improvements, as shown in Figs. 4 and 8 in Appendix. The more office building energy saved, the greater number of EVs can be supported at the offices parking lot and their charging demands can be satisfied. We applied the Bass model (Bass, 1969; Becker et al., 2009; Jensen et al., 2017; Liu and Kontou, 2022) to infer the corresponding year when we expect the EV adoption to grow to the energy levels dictated by the building energy savings adjustment and controls; this modeling outcome presented in Table 4. For example, if the HVAC setpoint in a Chicago, IL building office changes from 24 to 24.5 °C, the saved energy (47 kWh) can support roundtrip commuting charging demand of seven EVs, which is aligned with the EV adoption trend expected to be observed in the year 2024.

We consider AC Level 2 chargers with 6.6 kW maximum charging power in this paper. The electricity price schemes in Chicago IL, Baltimore MD, and Houston TX are obtained from local electric utilities, shown in Table 5 (ComEd, 2022; US Utility Rate Database, 2022b,a). Due to data limitations and an electricity tariff without a demand charge in Chicago, we compare the total costs including energy costs and demand charge, only in the cities of Baltimore and Houston, while we make comparisons between unmanaged and managed charging schedules for all three US cities.

3.2. Results with office cooling setpoint adjustments

We conducted numerical experiments using one hundred synthetic data samples and one typical sample of parameters with HVAC cooling setpoint adjustments on a typical summer workday. Fig. 5 presents the building load, EV charging profile, and "HVAC + EV load" with +0.5 and +2 °C cooling setpoint adjustments. Comparing the unmanaged "HVAC + EV load" and the optimized one, the peak load distinction is not significant when the setpoint adjustment is +0.5 °C; the maximum difference are the same with 13.2 kW in Chicago IL, Baltimore MD, and Houston TX. However, with increasing energy savings from the building setpoint adjustments of

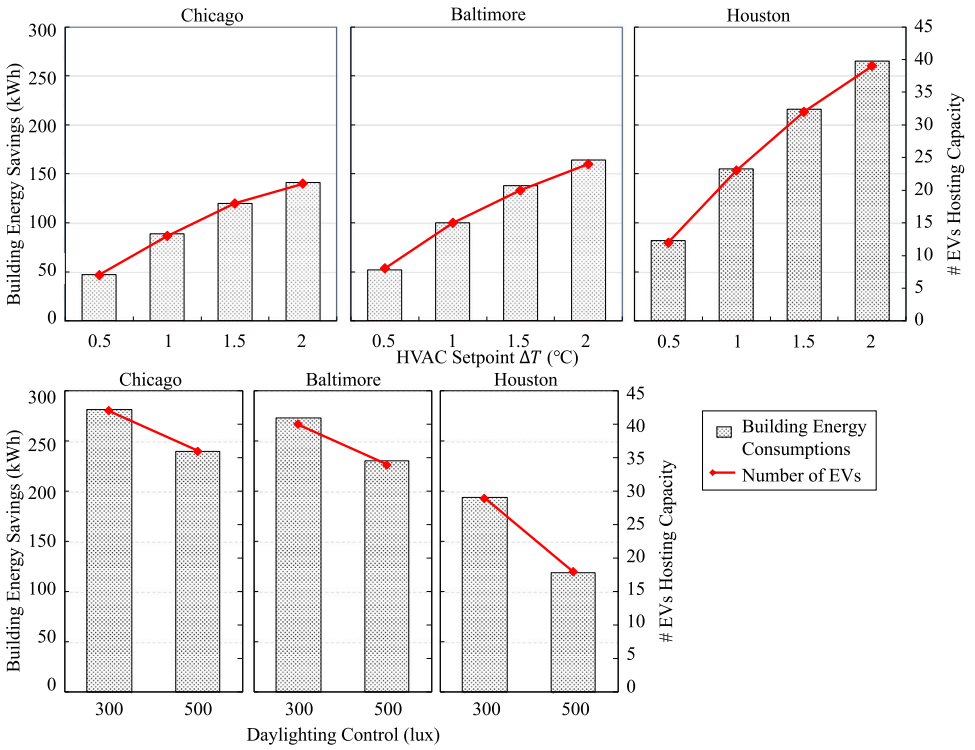


Fig. 4. Building energy savings from HVAC thermostat setpoint adjustments and daylighting control in the summer, and the corresponding number of EVs that can be supported by the saved energy in Chicago IL, Baltimore MD, and Houston TX, based on typical commuting patterns.

Table 5
Seasonal time-of-use electricity pricing applicable to office buildings in Chicago IL, Baltimore MD, and Houston TX.

		Electricity price (¢/kWh)		Demand charge ($/kW)	
		Summer	Winter	Summer	Winter
Chicago IL	10 pm–6 am	5.052	3.681	-	-
	6 am–2 pm & 7 pm–10 pm	9.653	5.247		
	2 pm–7 pm	14.966	15.865		
Baltimore MD	10 pm–6 am	5.854	5.854	3.84	3.84
	7 am–9 am & 8 pm–10pm	6.632	9.795		
	9 am–8 pm	9.795	6.632		
Houston TX	8 pm–12 pm	1.682	1.682	8.17	3.44
	12 pm–8 pm	4.882	1.943		

+2 °C that support more EVs to charge at the workplace, the peak load differences between the unmanaged and managed profile are a maximum of 39.6 kW in Chicago IL, 59.4 kW in Baltimore MD, and 71.4 kW in Houston TX. The peaks of unmanaged building load in the +2 °C scenarios are at least 1.5 times the building load in the baseline scenarios, which would burden the building manager or the corporation with significant demand charges, demonstrating the benefits of WPC management.

In Chicago IL, we compared the volumetric portion of the electricity costs before and after implementing the workplace EV charging management in the office's parking lot. We found that the total cost difference is rising (Fig. 6). Moreover, the daily electricity costs with managed EV charging and HVAC setpoint adjustments are even lower than the ones without EV charging (i.e., the baseline). This means that the cost savings from the building setpoint adjustments can, on average, cover the costs of EV charging. With the increased energy that can be saved from setpoint adjustments from +1 to +2 °C, the number of EVs supported can increase without significantly changing the total energy use in kWh, while charging management becomes essential for achieving further reductions in the electricity costs. Therefore, the average volumetric energy charge of the building and managed EV charging for the +2 °C scenario is less than the +1 and +1.5 °C scenarios. Comparing the office building and EV charging costs, including both the energy and demand charges in Baltimore MD and Houston TX (panels 2 and 3 in Fig. 6), the cost gap between unmanaged and managed plans is increasing. In Baltimore MD, with an increasing number of EVs in the office parking lot, the total costs present an upward trend for both the managed and unmanaged plans. In contrast, the daily total costs of the managed plan in Houston TX decrease from $104.81 in the +1 °C scenario to $102.58 in the +1.5 °C scenario, as well as the costs of $104.38 in the +2 °C

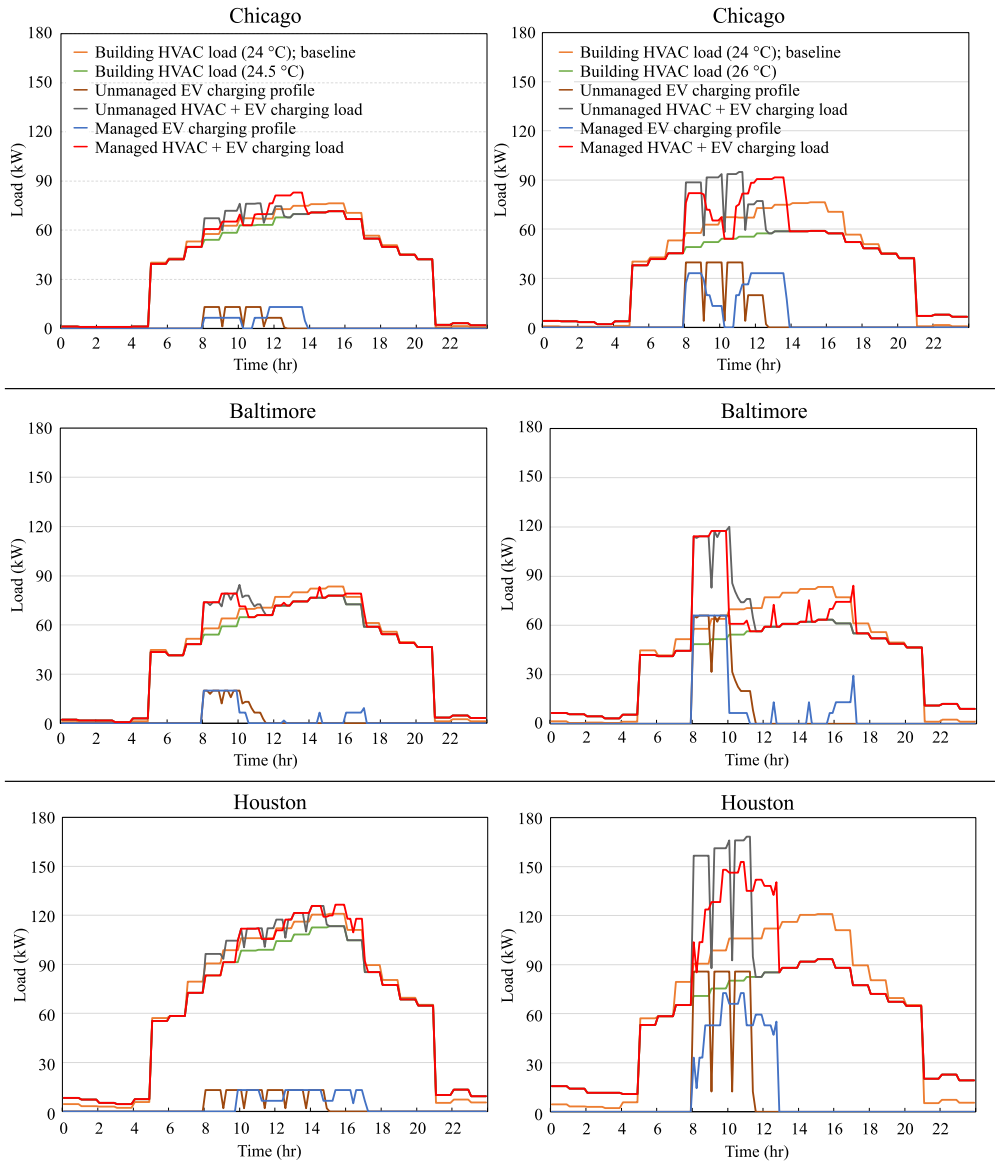


Fig. 5. Building load and EV workplace charging profiles under unmanaged and managed charging schedules with HVAC cooling setpoint adjustments (i.e., +0.5 °C and +2 °C) in Chicago IL, Baltimore MD, and Houston TX. The number of EVs in these scenarios are 7 for the +0.5 °C HVAC adjustment and 21 for the +2 °C HVAC adjustment in Chicago, 8 and 24 in Baltimore, and 12 and 39 in Houston, respectively.

Table 6
Relationship between the ratio of EVs to chargers and the building electricity and EV charging costs, including demand charges, of typical samples in Baltimore MD and Houston TX.

	Baltimore MD				Houston TX			
	+0.5 °C	+1 °C	+1.5 °C	+2 °C	+0.5 °C	+1 °C	+1.5 °C	+2 °C
Ratio (#EVs/#Chargers)	2.67	2.50	2.50	2.00	4.00	2.875	3.2	3.00
Daily Total Costs ($)	103.67	108.44	107.73	109.68	93.75	104.81	102.58	104.38

scenario. This is mainly driven by the demand charge, which is impacted by the ratio of the number of EVs to the number of chargers, presented in Table 6. The ratio in the +1.5 °C scenario is 3.2 and is larger than the ratios in the +1 and +2 °C scenarios, which are 2.875 and 3, respectively. This suggests that fewer EVs can charge at the same time in the +1.5 °C scenario, which results in lower load peaks and demand charges.

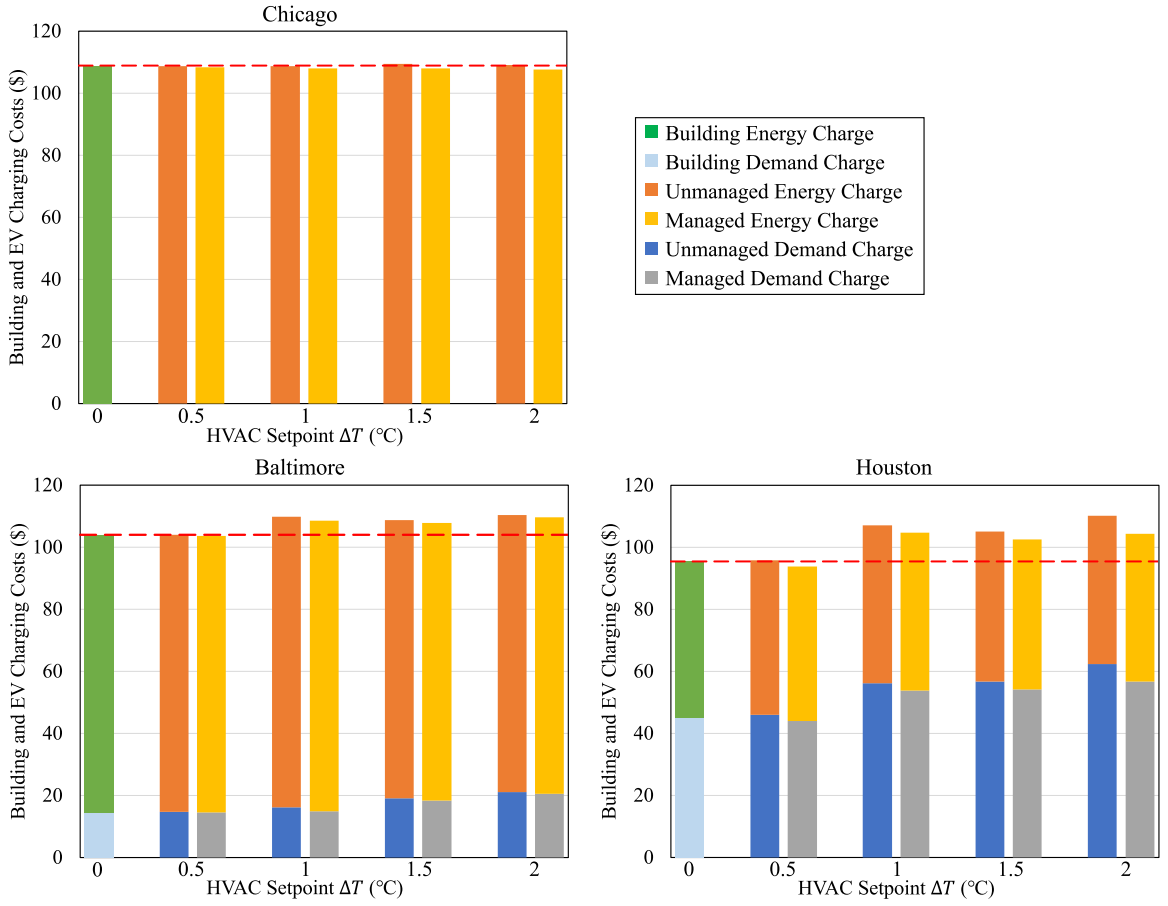


Fig. 6. Building and EV daily charging costs of typical samples with the consideration of demand charge for the cooling setpoint adjustment scenarios in Chicago IL, Baltimore MD, and Houston TX.

Fig. 7 presents the comparison of synthetic data samples' savings with the HVAC setpoint adjustments. Fig. 7(a) showcases the daily charging savings per day which only consider the volumetric energy charge of the electricity costs. Fig. 7(b) illustrates the building and EV charging cost savings per day, including both volumetric and demand charges in Baltimore MD and Houston TX. In Chicago IL, the average daily savings increase from $0.16 in the +0.5 °C scenario to $0.48 in the +2 °C scenario when more EVs charge in the office parking lot. The range of daily savings also expands from $0.59 in the +0.5 °C scenario to $1.49 in the +2 °C scenario. The average charging savings in +0.5 °C scenarios in Baltimore MD and Houston TX are $0.07 and $0.06, respectively; while in +2 °C scenarios, the average savings are $0.22 and $0.15, respectively. Compared to the increasing trend of charging savings in Chicago IL, from $0.16 to $0.48, the rising trends of charging savings in Baltimore MD and Houston TX are less significant. This is mainly driven by the difference of time-of-use electricity price schemes and the travel patterns (i.e., arrival and departure times) among Chicago IL, Baltimore MD, and Houston TX. The electricity prices during daytime are 9.653 ¢/kWh from 6 am to 2 pm and 14 ¢/kWh from 2 pm to 7 pm in Chicago IL. In Baltimore MD, the daytime electricity prices are 6.632 ¢/kWh from 7 am to 9 am and 9.795 ¢/kWh from 10 am to 7 pm. The electricity tariff tiers' differences in Baltimore MD is 3.163 ¢/kWh and is less than that of Chicago's. Similarly, Houston's electricity price difference between tiers is 3.2 ¢/kWh, which is smaller than that of Chicago. Based on the typical arrival and departure times in Chicago IL, Baltimore MD, and Houston TX, which are 8 am and 5 pm, respectively, the coincident periods between the EVs' dwell time and lower electricity rate times are 7 h, 2 h, and 5 h, for each city respectively. The longer coincident period in Chicago IL enables greater opportunities to complete more charging tasks during these lower electricity price periods with relatively lower rates than Baltimore MD and Houston TX.

When the demand charges are considered (Fig. 7(b)) in Baltimore and Houston, the saving ranges of the building and charging electricity costs with demand charge increase due to a greater volume of saved energy (given the HVAC setpoint adjustment, which increases the number of EVs that can recharge in the workplace). In Baltimore, the saving range rises from $4.27 in the +0.5 °C scenario to $8.25 in the +2 °C scenario, while the saving range increases from $11.12 to $28.01 in Houston. However, the median total cost savings, including volumetric energy charge and demand charge in Baltimore, fluctuate around 0, while the median total cost savings in Houston are negative. Compared to the typical samples shown in Fig. 6, in which the cost savings are positive in all HVAC setpoint adjustment scenarios, the travel patterns of the synthetic samples are diverse, which makes the range of cost savings

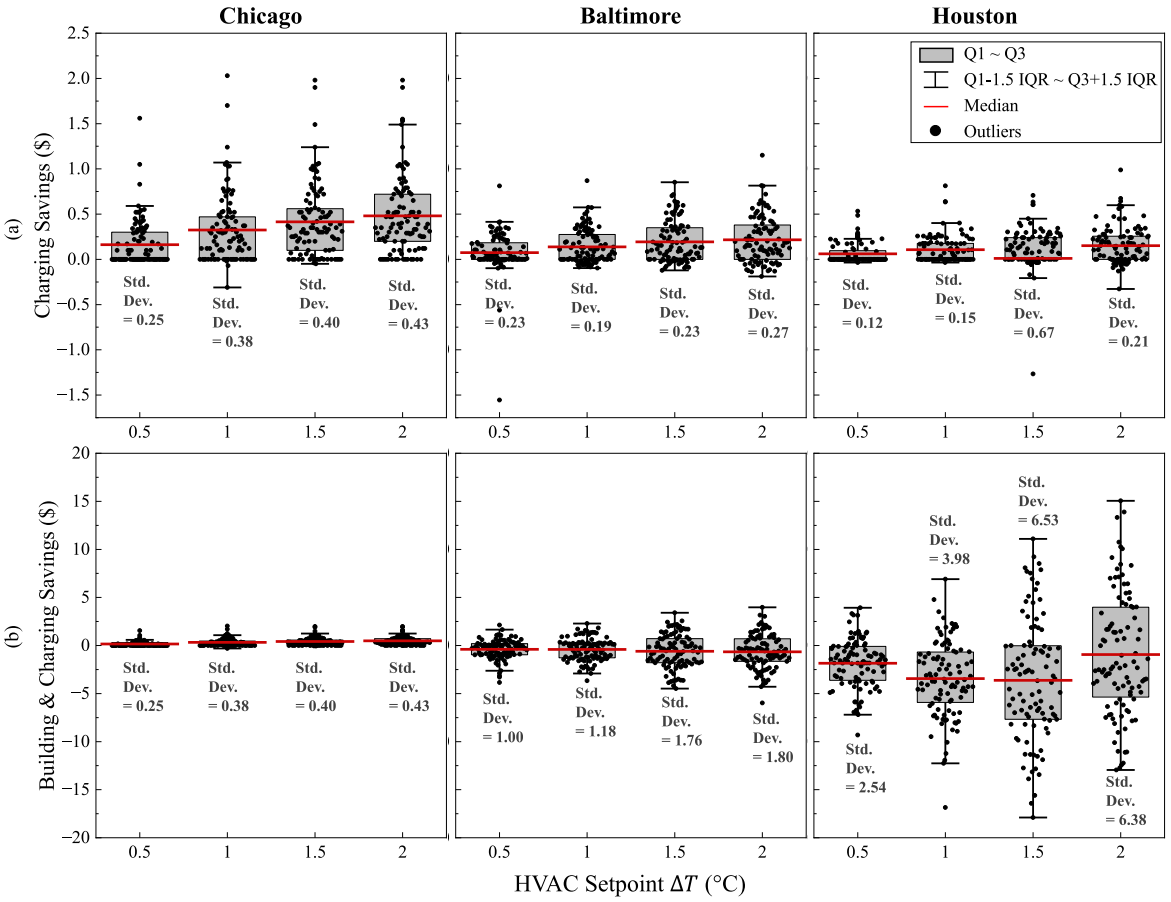


Fig. 7. Panel (a) shows the daily charging savings excluding demand charge, and panel (b) shows the daily savings of building and EV charging costs including energy charge and demand charge of synthetic data samples with the HVAC cooling setpoint adjustments in Chicago IL, Baltimore MD, and Houston TX.

wider, including positive and negative values. Therefore, the time-of-use electricity scheme, travel patterns, and demand charge can influence the performance of managed charging schedules compared to unmanaged ones.

3.3. Comparison of results with office cooling and heating setpoint adjustments

With heating setpoint adjustments, we implemented managed EV charging and the FCFS unmanaged approach and compared the management performance in one hundred synthetic data samples and a typical sample for each simulation scenario and city, respectively. Comparing Figs. 6 and 9, the daily electricity costs almost remains constant when implementing HVAC cooling or heating setpoint adjustments and enabling EV charging, considering the volumetric/energy charge in Chicago IL. Managed EV charging profiles in all scenarios can decrease the volumetric electricity costs compared to unmanaged ones, saving $1.43 in −0.5 °C scenario, $2.27 in −1 °C scenario, $3.81 in −1.5 °C scenario, and $6.19 in −2 °C scenario. The WPC management is more effective during winter in Chicago, as expected. For instance, the ratio of savings to the unmanaged electricity bills is 1.31% and 4.31% in +2 °C and −2 °C, respectively. In Baltimore MD and Houston TX, more distinct seasonal electricity bills differences can be found. Managed EV charging in the office building can slightly increase the electricity bills in summer, compared to the baseline electricity bills. Often with more EVs in the office parking lot, the burden to the building manager also presents an increasing trend. However, EV charging would not impose increased economic burden to the manager during the winter under the managed charging profile, when an even greater number of EVs are hosted, due to a more efficient building HVAC setting strategy that can reduce electricity bills. The ratio of electricity bill savings is compared between the FCFS unmanaged charging practice and the seasonal performance of the managed charging profile, as shown in Table 7. As expected, managed WPC performs better during winter in Baltimore MD, with a ratio of 0.62% for the +2 °C control strategy and a ratio of 10.43% for the −2 °C. However, the WPC management showcases greater economic benefits in Houston TX with 5.22% in the +2 ° scenario, comparing to the 2.61% in the −2 °C one.

In terms of the synthetic data samples, Table 8 presents the monthly average cost savings and cost savings per kWh with HVAC thermostat setpoint adjustments, to compare the management performance during winter and summer. We use the average cost savings to compare the seasonal WPC management performance in each city. The coordinated management of building efficiency

Table 7
Comparison of the ratio of electricity bill savings to a FCFS unmanaged charging profile of the typical samples during summer and winter.

HVAC setpoint ΔT (°C)	Chicago IL		Baltimore MD		Houston TX	
	Summer	Winter	Summer	Winter	Summer	Winter
+/− 0.5	0.33%	0.95%	0.30%	1.28%	2.12%	0.24%
+/− 1	0.66%	1.54%	1.15%	2.78%	2.18%	0.48%
+/− 1.5	1.30%	2.63%	0.80%	4.86%	2.33%	0.58%
+/− 2	1.31%	4.31%	0.62%	10.43%	5.22%	2.61%

* In Chicago IL, the cost savings include the volumetric charge of building and EV charging energy consumption; in Baltimore MD and Houston TX, the cost savings include the volumetric and demand charge of building and EV charging energy consumption.

Table 8
Seasonal comparison of monthly average cost savings and average cost savings per kWh of the synthetic data samples in Chicago IL, Baltimore MD, and Houston TX with HVAC thermostat setpoint adjustments.

	Summer				Winter			
	+0.5 °C	+1 °C	+1.5 °C	+2 °C	−0.5 °C	−1 °C	−1.5 °C	−2 °C
	Chicago IL							
Average Cost Savings ($)	3.52	7.04	9.02	10.56	4.60	8.40	13.20	17.60
Average Cost Savings per EV ($)	0.50	0.54	0.50	0.50	0.38	0.38	0.41	0.42
Average Cost Savings per kWh (cents)	7.70	8.14	7.70	7.48	6.20	5.80	6.20	6.40
	Baltimore MD							
Average Cost Savings ($)	−8.80	−8.80	−13.2	−14.52	28.40	71.60	143.00	213.00
Average Cost Savings per EV ($)	−1.10	−0.59	−0.66	−0.61	2.58	3.41	4.61	5.33
Average Cost Savings per kWh (cents)	−16.94	−8.80	−9.46	−8.80	38.80	50.60	69.40	80.20
	Houston TX							
Average Cost Savings ($)	−40.48	−75.68	−79.86	−20.68	21.6	39.6	69.4	99.8
Average Cost Savings per EV ($)	−3.37	−3.29	−2.50	−0.53	2.40	2.33	3.02	3.56
Average Cost Savings per kWh (cents)	−49.5	−48.84	−36.96	−7.7	36.6	36	45.2	52.6

* In Chicago IL, the cost savings include the volumetric charge of building and EV charging energy consumption; in Baltimore MD and Houston TX, the cost savings include the volumetric and demand charge of building and EV charging energy consumption.

and charging schedule in Chicago IL performs better during winter, saving $4.6 in −0.5 °C and $17.6 in −2 °C, which is aligned with the results of typical cases. More energy is saved after the HVAC setpoint adjustments are implemented in the winter. In Baltimore MD and Houston TX, cost savings include volumetric and demand charge ones for the coupled building and WPC energy consumption. The management performs better during winter both in Baltimore and Houston. In contrast, the average cost savings in summer are negative, which means that the managed charging schedule can even result in higher electricity bills because of the higher demand charge due to the EV charging load. Moreover, the coordinated management is better performing in Baltimore, due to a larger gap of time-of-use electricity price tiers. In the synthetic samples, the system's management performance in Houston is not aligned with the results of the typical samples. This is mainly because of the travel patterns distribution, which could result in an expensive demand charge. To compare the management performance across the three representative cities, we use the average cost savings per kWh. For example, the monthly winter average cost savings per kWh increases from 38.8 ¢ to 80.2 ¢ in Baltimore, while in Houston these rise from 36.6 ¢ to 22.6 ¢. Thus, the management is more effective in Baltimore during winter.

3.4. Results with office daylighting control

Similarly, we constructed one hundred synthetic data samples and one typical sample to compare the office building and EV WPC costs under unmanaged (FCFS) and managed charging schedules for different daylighting control scenarios during summer and winter. Table 9 illustrates the number of EVs, monthly charging savings, average monthly charging savings of synthetic samples, monthly charging savings per EV, and ratio of charging savings to the FCFS costs in synthetic data and typical samples in the three US cities during summer, while Table 10 presents the results during winter. For the typical samples in Table 9, managed charging performs better in the 300 lux scenarios in Chicago, Baltimore, and Houston than in the 500 lux scenarios, as expected. The monthly charging savings in the 300 lux scenario in Chicago are $78.45, larger than the charging savings of $60.70 in the 500 lux scenario. The ratio of savings compared to FCFS charging costs in Houston is significantly larger than Chicago and Baltimore, mainly due to the relatively smaller magnitude of charging costs. In Houston's 300 lux scenario, the daily charging costs of managed and unmanaged schedules are $3.3 and $5.1, respectively. However, the daily charging costs of managed and unmanaged schedules are $27.2 and $30.8 in Chicago, which makes its ratio of charging savings relatively smaller than Houston's. For the synthetic data samples, the ranges of charging savings, charging savings per EV, and the ratio of savings to FCFS costs are wider in the 300 lux scenarios than in the 500 lux scenarios in all three cities. Fig. 10 presents the daily office building and charging savings, and building and

Table 9
Comparison of the managed charging with daylighting control strategies of typical and synthetic data samples in Chicago IL, Baltimore MD, and Houston TX during summer.

Typical samples		Number of EVs	Monthly charging savings ($)	Monthly charging savings per EV (¢/veh)	Savings/FCFS costs (%)	
Chicago IL	300 lux	42	78.45	186.76	11.59	
	500 lux	36	60.70	168.59	10.54	
Baltimore MD	300 lux	39	44.07	112.97	9.70	
	500 lux	33	22.26	67.43	6.07	
Houston TX	300 lux	29	40.50	139.68	35.83	
	500 lux	18	22.24	123.60	33.07	
Synthetic data samples		Number of EVs	Monthly charging savings ($)	Average monthly savings ($)	Monthly charging savings per EV (¢/veh)	Savings/FCFS costs (%)
Chicago IL	300 lux	32–43	−1.33 to 75.87	19.43	−3.51 to 210.75	−0.220 to 11.955
	500 lux	27–39	−2.43 to 52.33	17.20	−7.85 to 161.63	−0.471 to 9.785
Baltimore MD	300 lux	33–44	−8.83 to 24.13	5.77	−22.07 to 65.22	−1.987 to 5.061
	500 lux	27–39	−3.84 to 23.79	6.33	−11.79 to 72.94	−1.017 to 5.876
Houston TX	300 lux	22–33	−4.16 to 17.63	1.94	−15.99 to 67.80	−4.544 to 15.272
	500 lux	14–21	−2.07 to 11.72	1.57	−13.77 to 66.84	−3.855 to 18.123

Table 10
Comparison of the managed charging with daylighting control strategies of typical and synthetic data samples in Chicago IL, Baltimore MD, and Houston TX during winter.

Typical samples		Number of EVs	Monthly charging savings ($)	Monthly charging savings per EV (¢/veh)	Savings/FCFS costs (%)	
Chicago IL	300 lux	26	52.34	2.02	22.23	
	500 lux	22	26.16	1.18	14.45	
Baltimore MD	300 lux	25	72.56	2.90	23.66	
	500 lux	21	74.24	3.54	27.41	
Houston TX	300 lux	22	3.98	0.18	7.38	
	500 lux	16	3.76	0.24	9.36	
Synthetic data samples		Number of EVs	Monthly charging savings ($)	Average monthly savings ($)	Monthly charging savings per EV (¢/veh)	Savings/FCFS costs (%)
Chicago IL	300 lux	21–31	−2.34 to 39.50	9.30	−10.28 to 197.52	−1.185 to 16.813
	500 lux	17–24	−1.01 to 38.95	8.54	−5.29 to 238.05	−0.488 to 19.205
Baltimore MD	300 lux	20–29	49.97 to 98.51	71.07	229.04 to 516.02	16.545 to 29.737
	500 lux	17–24	32.31 to 86.00	57.55	177.71 to 516.93	14.382 to 31.326
Houston TX	300 lux	16–24	2.94 to 6.47	4.64	14.08 to 35.47	5.618 to 11.621
	500 lux	11–20	−6.37 to 4.94	3.11	−58.42 to 34.54	1.184 to 12.131

charging savings per kWh of the synthetic data samples. In Chicago, there is no demand charge, thus, the daily savings magnitude is approximately $0.5. However, in Baltimore, the managed charging profile performs better in winter, saving on average 1.53 ¢/veh in winter and −0.40 ¢/veh in summer with a 300 lux daylighting setting; while WPC management has a more effective performance during summer in Houston, saving on average 1.72 ¢/veh in winter and 2.20 ¢/veh in the summer.

4. Conclusion

In this paper, we propose a framework to manage the EV charging schedule at the workplace, coupled with HVAC setpoint adjustments and daylighting control in a medium office building. Taking Chicago IL, Baltimore MD, and Houston TX as representative cities of three different climate regions in the US, we simulate the energy consumption of the office building under different energy control scenarios. The energy savings of the adjusted HVAC setpoint and lighting setting scenarios compared to a baseline are

obtained in these three cities. Based on commuting travel patterns (i.e., workplace arrival and departure times, charging demands), we estimate the number of EVs that can be supported by the building energy saved. We developed an optimization model aiming to minimize the charging costs of the EVs colocated with the office building, considering a limited number of chargers at the workplaces parking and that charging stations are shared among the EV commuters.

We implement the proposed coupled management framework in Chicago IL, Baltimore MD, and Houston TX, and the main findings of this study are as follows:

- The climate of each city in the present study influences the amount of energy savings from HVAC setpoint adjustments at the office building, as expected. Higher outdoor air temperatures in the summer result in higher energy savings by each adjustment step of the cooling setpoint. In contrast, lower outdoor air temperatures in the winter result in higher energy savings by each adjustment step of the heating setpoint. The latitude of the cities can impact the energy savings from daylighting control. The higher the latitude of the city, the greater energy savings are yielded with the same daylighting control scheme.
- Applying the optimization model to manage the workplace charging schedule in the typical samples, we ensure that the managed EV charging will not result in significantly higher total energy costs with or without the consideration of demand charge, compared to an unmanaged FCFS EV charging schedule.
- In synthetic data samples, the cost-saving performance of the managed charging schedule is influenced by the model inputs (e.g., time-of-use electricity price and commuting travel patterns). The larger price gap between different times of the electricity tariff tiers can increase the charging savings in the managed EV charging schedule; the longer coincident period of low electricity prices and dwell times of commuters at the workplace parking lot results in managed charging schedules that are more cost-effective.

This research is introducing opportunities for coupled energy efficiency and economic gains at the EV-office building nexus. Future research in this space can address this study's limitations. We refrained from determining the optimal number of chargers in the medium office building, a process that needs to consider not only the charging demand, but also the budget for the charger installation and operation, the business model of workplace charging, charging waiting time, etc. Looking into the synthetic samples results, we found that the number of chargers affect the charging profile, and the ratio of EVs to chargers affects the electricity bills thus, the impact of different EVs to chargers ratio should be further analyzed in the near future. The building control strategies are implemented separately in this research; however, combining them could significantly improve building efficiency and meet increased EV charging requirements without augmenting total energy consumption. The combination of office building energy efficiency control strategies including HVAC setpoint adjustments and daylighting control can be implemented with the WPC management to analyze the cost-saving performance of these practices in different kinds of buildings. Finally, a variety of charger power levels (e.g., Level 2 with low power, Level 2 with medium power, and direct-current fast charger) can be considered in the future research of EV charging in office buildings.

CRediT authorship contribution statement

Shanshan Liu: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Alex Vlachokostas:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Chen Si:** Data curation. **Eleftheria Kontou:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

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Appendix

See Figs. 8–10.

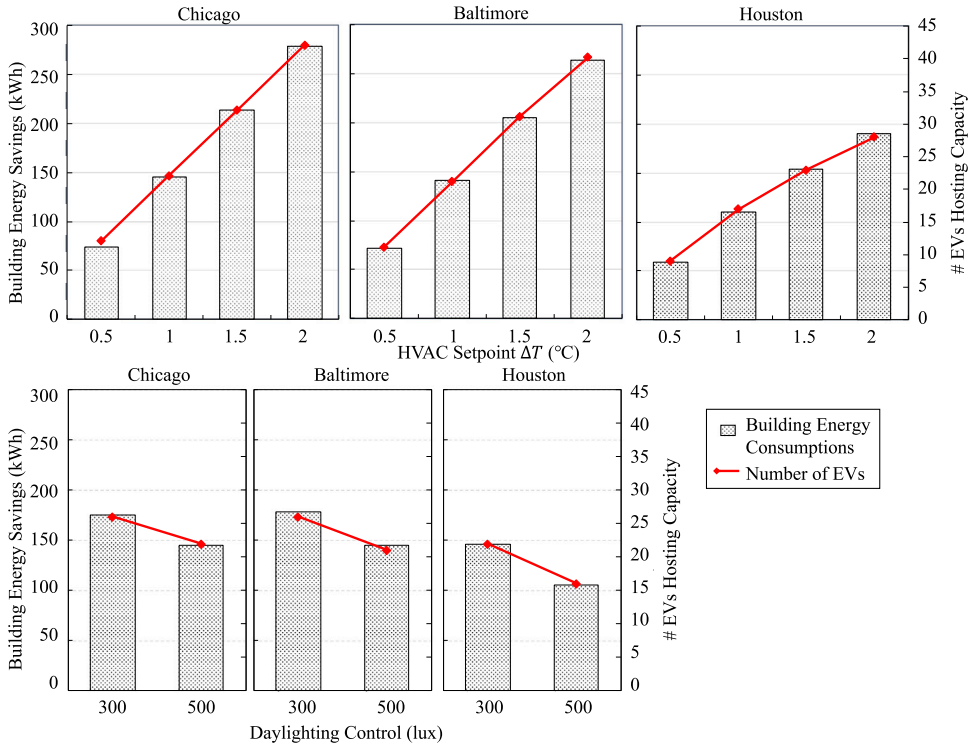


Fig. 8. Building energy savings from HVAC thermostat setpoint adjustments and daylighting control in winter, and the corresponding number of EVs that can be supported by the saved energy in a representative office building in Chicago IL, Baltimore MD, and Houston TX.

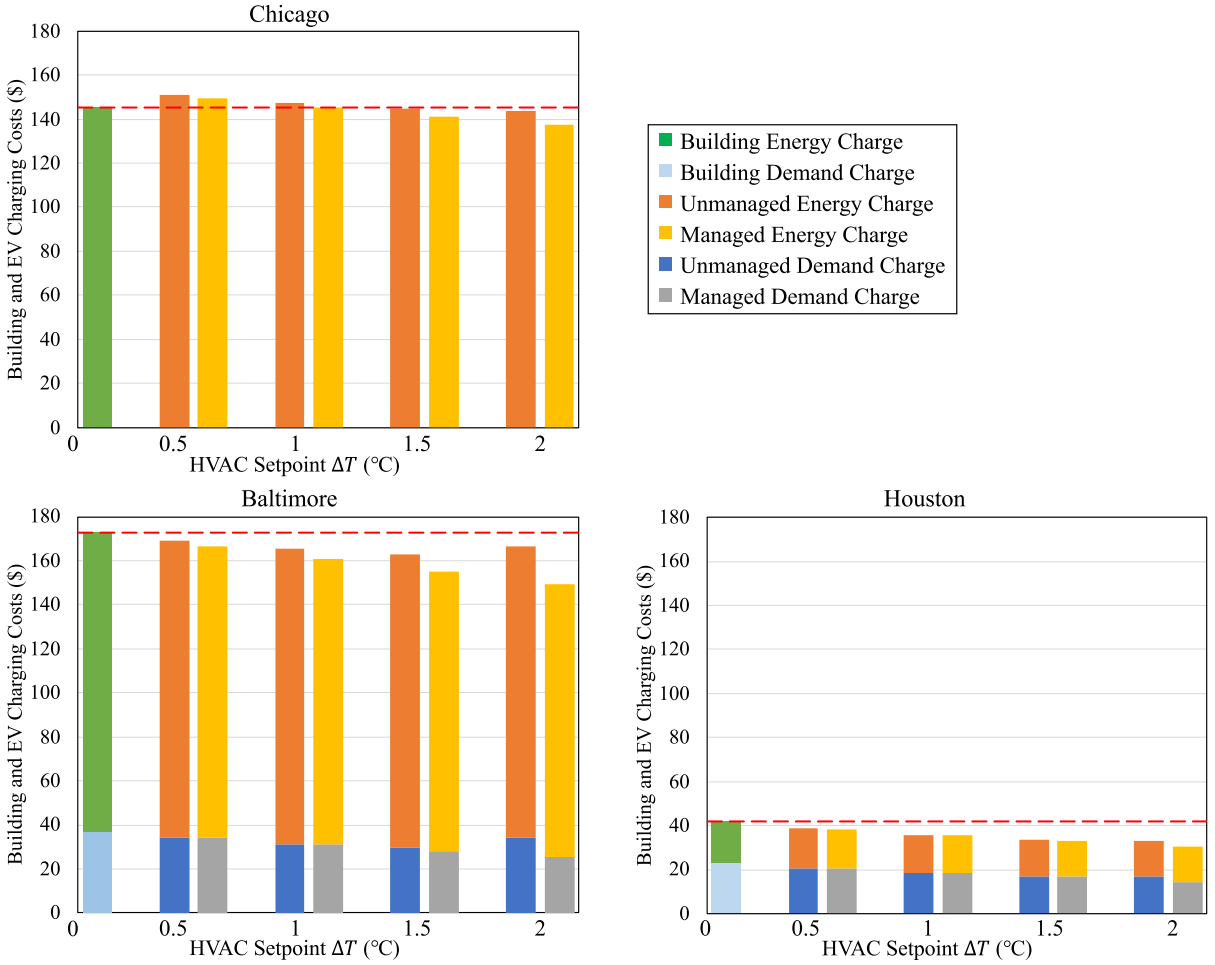


Fig. 9. Building and EV charging costs for HVAC heating setpoint adjustment scenarios. In Chicago IL, the costs only include volumetric/energy charge, while in Baltimore MD and Houston TX, the costs include volumetric/energy and demand charge.

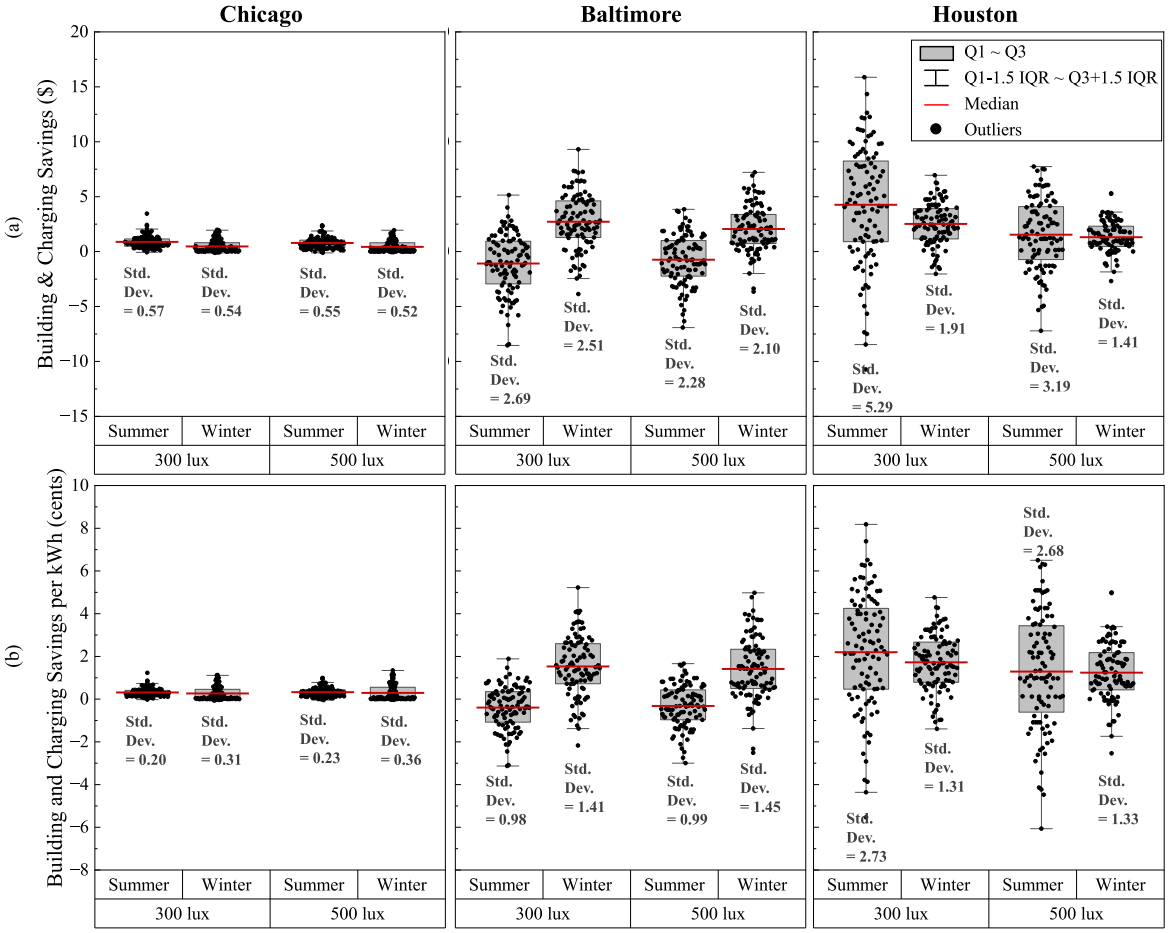


Fig. 10. Daily building and charging savings, and building and charging savings per kWh with daylighting control of synthetic data samples in Chicago IL, Baltimore MD, and Houston TX. In Chicago IL, the savings include volumetric charge, while they include both volumetric/energy and demand charges in Baltimore and Houston.

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